In the concrete example where kets are represented by column vectors, the dual bras are represented by row vectors, with the entries being drawn from the complex conjugate numbers. Thus, if the ket |*A*> is represented by the column

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{pmatrix},$$

then the corresponding bra <*A*| is represented by the row

$$\begin{pmatrix} \alpha_1^* & \alpha_2^* & \alpha_3^* & \alpha_4^* & \alpha_5^* \end{pmatrix}.$$

1.9.4 Inner Products

You are no doubt familiar with the dot product defined for ordinary 3-vectors. The analogous operation for bras and kets is the *inner product*. The inner product is always the product of a bra and a ket and it is written this way:

$$\langle B|A\rangle.$$

The result of this operation is a complex number. The axioms for inner products are not too hard to guess:

1. They are linear:

$$\langle C|\{\ |A\rangle + |B\rangle\ \} = \langle C|A\rangle + \langle C|B\rangle.$$